20 Verified Scholarly Papers: Detecting Misinterpreted but Genuine Citations through Automated Claim Verification

This list focuses on citations that *exist and point to a real source* but whose content is mischaracterized, oversimplified, contradicted, or otherwise misrepresented by the citing text (as opposed to fabricated/hallucinated citations, which are included only where directly relevant to the verification methods used).

Survey and Review Papers

Quotation accuracy in medical journal articles — a systematic review and meta-analysis Hannah Jergas, Christopher Baethge, 2015, *PeerJ* Paper / DOI Landmark meta-analysis of 28 studies synthesizing quotation/citation error rates (~25%) across medical fields; the classification of major vs. minor quotation errors underpins most later automated detection schemes.

Accuracy of cited "facts" in medical research articles: A review of study methodology and recalculation of quotation error rate Scott A. Mogull, 2017, *PLOS ONE* Paper / DOI Reviews methodological inconsistencies across quotation-accuracy studies and recalculates a pooled error rate, providing a methodological baseline for what "misrepresentation" means in citation-checking research.

How do authors perceive the way their work is cited? Findings from a large-scale survey on quotation accuracy Simon Wakeling, Monica Lestari Paramita, Stephen Pinfield, 2025, *Journal of the Association for Information Science and Technology (JASIST)* Paper / DOI Recent survey synthesizing 23 quotation-accuracy studies and directly surveying cited authors about misrepresentation, bridging manual citation-accuracy research and computational detection needs.

Quotation errors in general science journals Neal Smith Jr, Aaron Cumberledge, 2020, *Proceedings of the Royal Society A* Paper / DOI Reviews the long-running cross-disciplinary literature on quotation/citation error rates and situates the "citation vs. quotation accuracy" distinction that automated systems must operationalize.

Foundational Papers

How citation distortions create unfounded authority: analysis of a citation network Steven A. Greenberg, 2009, *BMJ* Paper / DOI Foundational study formalizing "citation distortion" (bias, amplification, invention) — genuine citations that collectively misrepresent a claim's evidentiary support — and a conceptual ancestor of automated misrepresentation detection.

FEVER: a Large-scale Dataset for Fact Extraction and VERification James Thorne, Andreas Vlachos, Christos Christodoulopoulos, Arpit Mittal, 2018, *NAACL-HLT* Paper / DOI Foundational general-domain claim verification dataset/task (SUPPORTS/REFUTES/NOT ENOUGH INFO) whose pipeline architecture (retrieval → evidence selection → entailment) underlies nearly all later scientific claim/citation verification systems.

Argumentative Zoning: Information Extraction from Scientific Text Simone Teufel, 1999, PhD Thesis, University of Edinburgh Paper / DOI Establishes the rhetorical-status framework for analyzing what a citing sentence is actually claiming about a source, foundational to later work distinguishing accurate from misleading citation function.

Automatic Classification of Citation Function Simone Teufel, Advaith Siddharthan, Dan Tidhar, 2006, *EMNLP* Paper / DOI First large-scale automated classifier of *why* a citation is made (e.g., contrast, basis, weakness); a direct precursor to systems that must first understand citation intent before judging whether the citation misrepresents its source.

Recent Research Papers

Assessing citation integrity in biomedical publications: corpus annotation and NLP models Maria Janina Sarol, Shufan Ming, Shruthan Radhakrishna, Jodi Schneider, Halil Kilicoglu, 2024, *Bioinformatics* (Oxford Academic) Paper / DOI Directly on-topic: builds a 3,063-instance annotated corpus of genuine biomedical citations labeled ACCURATE/NOT_ACCURATE/IRRELEVANT with fine-grained error types (misquote, oversimplify, contradict), and trains claim-verification NLP models (BM25+MonoT5 retrieval + MultiVerS) to detect misrepresented-but-real citations.

SemanticCite: Citation Verification with AI-Powered Full-Text Analysis (2025/2026), *arXiv preprint* Paper / DOI Proposes deep full-text semantic analysis (vs. abstract-level/binary approaches) to identify subtle, "partially supported" citation issues — genuine citations whose claims are only weakly backed by the source.

CiteAudit: You Cited It, But Did You Read It? A Benchmark for Verifying Scientific References in the LLM Era (2026), *arXiv preprint* Paper / DOI Introduces a multi-agent (Extractor, Memory, Web Search, Scholar, Judge) framework and benchmark for verifying reference existence and metadata/content consistency at scale.

CiteCheck: Retrieval-Grounded Detection of LLM Citation Hallucinations in Scientific Text (2026), *arXiv preprint* Paper / DOI Frames citation checking as grounded retrieval plus structured metadata/content comparison; its "minor hallucination" class (real paper, perturbed/misrepresented content) closely maps to the misinterpreted-but-genuine citation problem.

HalluCiteChecker: A Lightweight Toolkit for Hallucinated Citation Detection and Verification in the Era of AI Scientists (2026), *arXiv preprint* Paper / DOI Formalizes automated citation verification as an NLP task and delivers an offline, CPU-only toolkit intended for pre-submission and peer-review checks, distinguishing existence errors from content-support errors.

Methods / Algorithms

Fact or Fiction: Verifying Scientific Claims David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, Hannaneh Hajishirzi, 2020, *EMNLP* Paper / DOI Introduces the SciFact task/dataset and a three-stage pipeline (document retrieval → rationale sentence selection → SUPPORTS/REFUTES/NOINFO entailment) that is the core algorithmic template used to check whether a citation's claim is actually backed by its source.

Robust Claim Verification Through Fact Detection Nazanin Jafari, James Allan, 2024, *arXiv preprint* Paper / DOI Proposes FactDetect, which extracts and labels short factual statements from evidence to improve robustness and explainability of claim-verification models, directly applicable to pinpointing *why* a citation's claim diverges from its source.

Scientific Claim Verification with Fine-Tuned NLI Models Miloš Košprdić, Adela Ljajić, Darija Medvecki, Bojana Bašaragin, Nikola Milošević, 2024, *Proceedings of the 16th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management (IC3K 2024) — KMIS track* Paper / DOI Reframes claim–evidence relationship classification as textual entailment (NLI) using fine-tuned transformer models on SciFact, offering a lightweight methodological alternative for support/contradiction detection.

Retrieval Augmented Scientific Claim Verification Hao Liu, Ali Soroush, Jordan G. Nestor, Elizabeth Park, Betina Idnay, Yilu Fang, Jane Pan, Stan Liao, Marguerite Bernard, Yifan Peng, Chunhua Weng, 2024, *JAMIA Open* (Oxford Academic) Paper / DOI Combines retrieval augmentation with fine-tuned entailment models across FEVER, SciFact, and manually curated corpora to improve detection of claims not actually supported by their evidence — reducing the error propagation problem in earlier pipelines.

Applications

Quotation Accuracy Matters: An Examination of How an Influential Meta-Analysis on Active Learning Has Been Cited Amedee Marchand Martella, Jane Kinkus Yatcilla, Ronald C. Martella, et al., 2021, *Review of Educational Research* Paper / DOI Applied, large-scale manual audit (later informing automated approaches) showing over a third of citing articles made unsupported assertions about a single influential paper — a concrete real-world case of the problem automated tools aim to catch at scale.

Compound Deception in Elite Peer Review: A Failure Mode Taxonomy of 100 Fabricated Citations at NeurIPS 2025 Samar Ansari, 2026, *arXiv preprint* Paper / DOI Applied taxonomy of citation failures at a top-tier venue, arguing for mandatory automated verification at submission; useful for grounding claim-verification tools in a real peer-review application context.

Evaluation Methods / Benchmarks

SciFact-Open: Towards Open-Domain Scientific Claim Verification David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Iz Beltagy, Lucy Lu Wang, Hannaneh Hajishirzi, 2022, *Findings of EMNLP* Paper / DOI Extends SciFact to an open-domain retrieval setting over a large corpus, providing a harder, more realistic benchmark for whether a claim is genuinely supported, refuted, or unaddressed by candidate cited sources.

SciClaimHunt: A Large Dataset for Evidence-based Scientific Claim Verification (2025), *arXiv preprint* Paper / DOI Introduces a large-scale evidence-based scientific claim verification dataset built from a broader publicly available claim corpus, providing a distinct, larger-scale benchmark complementary to SciFact for training and evaluating support/refute classification models.

Note: This is an actively evolving research area (most items dated 2024–2026). Several arXiv preprints listed are not yet peer-reviewed; verify current publication status via the linked DOI/arXiv page before citing formally.